

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2352

North Campus Unm 100 kV
December 01, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2352. The Interconnection Customer proposes a 57.1 MW battery energy storage facility interconnecting at the North Campus Unm 100 kV substation in the Ohio area. The request was received on September 28, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 327 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$189,648,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SMITH MTN / HAVILAH 100.0 kV line	BEAVER / EUFAULA 500.0 kV	224.5	105	57.4
SMITH MOUNTAIN SUBSTATION / GREEN HILL 100.0 kV	BEAVER / EUFAULA 500.0 kV	185.6	116	57.4
WEST STATION NJ / HAVILAH 100.0 kV	OBERLIN OH / EMORY RD 100.0 kV	178.4	111	54.7
RED HILLS GENERATING FACILITY / WEST MEDINA SUBSTATION 100.0 kV	CANALROAD / NEWPORT TIE 230.0 kV L	178.0	128	17.6
EMORY ROAD SUBSTATION / BAYSIDE 100.0 kV line	NORTH CAMPUS UNM SUBSTATION / BAYSIDE 100.0 kV	176.8	120	33.1
NORT CAMPUS UNM / WEST STA NJ 100.0 kV	OBERLIN OH / EMORY RD 100.0 kV	176.5	120	54.7
VELCO HIGHGATE / OBERLIN OH 100.0 kV	N CAMPUS UNM / BAYSIDE 100.0 kV Line	175.1	118	33.1
VELCO HIGHGATE / HAVILAH 100.0 kV	BAYSIDE / N CAMPUS UNM 100.0 kV	174.3	115	33.1
RYAN / WEST MEDINA 100.0 kV Line	NEWPORT TIE / CANAL RD 230.0 kV	173.3	159	17.6
NORTECH / CITY OF SELBY ELECTRIC DISTRIBUTION 100.0 kV	NORTECH / BARRE CITY 100.0 kV	171.8	135	39.4
BAYSIDE SUBSTATION / NORTH CAMPUS UNM 100.0 kV L	OBERLIN OH / EMOY ROAD 100.0 kV	169.4	137	45.3
EORY ROAD / OBERLIN OH 100.0 kV	NORTH CAMPUS UNM / BAYSIDE 100.0 kV Line	161.3	136	33.1
BAYSIDE SUBSTATION / KNIGHT 100.0 kV	BEAVER / EUFAULA 500.0 kV Line	160.9	192	42.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CITY OF SHELBY ELECTRIC DISTRIBUTION SUBSTATION / PICKFORD 100.0 kV	BARRE CITY / NORTECH SUBSTATION 100.0 kV Line	156.9	147	39.4
HERITAGE / GREEN HILL 100.0 kV	BEAVER / EUFAULA SUBSTATION 500.0 kV Line	155.9	126	57.4
NEWPORT TIE / W SALEM 100.0 kV	CANAL RD / NEWPORTTIE 230.0 kV	150.9	143	9.8
NORTECH / KNIGHT 100.0 kV	BEAVER / EUFAULA 500.0 kV	150.5	213	42.6
RYAN / HERITAGE 100.0 kV	NORTECH / KNIGHT 100.0 kV	143.4	161	100.0
PICKFORD 100-1 kV Auto	ADAMS / ONANCOCK DIESEL GENERATION FACILITY SUBSTATION 230.0 kV L	136.9	458	19.0
NORTECH SUBSTATION / BARRE CITY 100.0 kV	PICKFORD 230-1 kV XFMR	129.9	188	19.2
PICKFORD 230/1kV TX	ONANCOCK DIESEL GENERATION FACILITY SUBSTATION / ADAMS 230.0 kV	127.0	509	19.0
ACME HILL / SEVILLE 100.0 kV line	NEWPORT TIE / CANAL RD 230.0 kV	124.5	183	12.2
RYAN / SEVILLE 100.0 kV	CANAL RD SUBSTATION / NEWPORT TIE 230.0 kV	124.5	183	12.2
WEST SALEM / ACME HILL 100.0 kV Line	CANAL D / NEWPORT TIE SUBSTATION 230.0 kV	122.9	192	12.2
BARRE CITY / OAKGROVE 100.0 kV	TROY ENERGY LLC / TREGO 500.0 kV line	120.0	132	6.2
SECOND FORK / PICKFORD 100.0 kV	ONANCOCK DIESEL GENERATION FACILITY / ADAMS 230.0 kV L	118.1	211	5.2
RYAN 100 kV/1 kV TX	NORTECH / KNIGHT 100.0 kV line	110.2	178	47.4
RYAN 230 kV/1 kV TX	NORTECH / KNIHT 100.0 kV	108.1	181	47.4
11TH AVENUE N / SOUTH BERWICK 100.0 kV Line	BARRE CITY SUBSTATION / OAKROVE 100.0 kV	106.1	142	7.3
PICKFORD / BRONX 100.0 kV	SECOND FORK / PICKFORD 100.0 kV Line	105.4	171	5.6
SLOCOMB / MARSH 100.0 kV	BARRE CITY / OAKGROVE SUBSTATION 100.0 kV	102.3	126	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SMITH MTN / HAVILAH 100.0 kV	Rebuild line with higher-capacity conductor	\$5,194,000	100% (\$5,194,000)
SMITH MOUNTAIN / GREEN HILL 100.0 kV	Rebuild line with higher-capacity conductor	\$7,081,000	100% (\$7,081,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST STATION NJ / HAVILAH 100.0 kV	Rebuild line with higher-capacity conductor	\$4,630,000	100% (\$4,630,000)
RED HILLS GENERATING FACILITY / W MEDINA 100.0 kV	Rebuild line with higher-capacity conductor	\$3,689,000	100% (\$3,689,000)
BAYSIDE / EMORY RD 100.0 kV	Rebuild line with higher-capacity conductor	\$1,003,000	100% (\$1,003,000)
NORT CAMPUS UNM / WEST STA NJ 100.0 kV	Rebuild line with higher-capacity conductor	\$7,820,000	100% (\$7,820,000)
VELCO HIGHGATE / OBERLIN OH 100.0 kV	Rebuild line with higher-capacity conductor	\$3,286,000	100% (\$3,286,000)
HAVILAH / VELCO HIGHGATE 100.0 kV	Rebuild line with higher-capacity conductor	\$4,921,000	100% (\$4,921,000)
WEST MEDINA SUBSTATION / RYAN SUBSTATION 100.0 kV	Rebuild line with higher-capacity conductor	\$3,174,000	100% (\$3,174,000)
NORTECH / CITY OF SELBY ELECTRIC DISTRIBUTION 100.0 kV	Rebuild line with higher-capacity conductor	\$822,000	100% (\$822,000)
N CAMPUS UNM / BAYSIDE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,424,000	100% (\$2,424,000)
EMORY ROAD / OBERLIN OH 100.0 kV	Rebuild line with higher-capacity conductor	\$1,580,000	100% (\$1,580,000)
BAYSIDE SUBSTATION / KNIGHT 100.0 kV	Rebuild line with higher-capacity conductor	\$5,961,000	100% (\$5,961,000)
CITY OF SHELBY ELECTRIC DISTRIBUTION SUBSTATION / PICKFORD 100.0 kV	Rebuild line with higher-capacity conductor	\$13,556,000	100% (\$13,556,000)
HERITAGE / GREEN HILL 100.0 kV	Rebuild line with higher-capacity conductor	\$6,197,000	100% (\$6,197,000)
WEST SALEM / NEWPORT TIE 100.0 kV Line	Rebuild line with higher-capacity conductor	\$6,759,000	100% (\$6,759,000)
NORTECH / KNIGHT 100.0 kV	Rebuild line with higher-capacity conductor	\$20,300,000	100% (\$20,300,000)
RYAN / HERITAGE 100.0 kV	Rebuild line with higher-capacity conductor	\$4,116,000	100% (\$4,116,000)
PICKFORD 100 kV/1 kV TX	Replace transformer with higher-rated unit	\$6,570,000	100% (\$6,570,000)
NORTECH SUBSTATION / BARRE CITY 100.0 kV	Rebuild line with higher-capacity conductor	\$23,243,000	100% (\$23,243,000)
PICKFORD 230-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$11,059,000	100% (\$11,059,000)
ACME HILL / SEVILLE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,555,000	40% (\$1,008,000)
RYAN / SEVILLE 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,598,000	40% (\$631,000)
WEST SALEM / ACME HILL 100.0 kV Line	Rebuild line with higher-capacity conductor	\$10,082,000	40% (\$3,978,000)
BARRE CITY / OAKGROVE 100.0 kV	Rebuild line with higher-capacity conductor	\$15,338,000	100% (\$15,338,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SECOND FORK / PICKFORD 100.0 kV	Rebuild line with higher-capacity conductor	\$5,139,000	100% (\$5,139,000)
RYAN 100 kV/1 kV TX	Replace transformer with higher-rated unit	\$3,953,000	86% (\$3,394,000)
RYAN 230 kV/1 kV TX	Replace transformer with higher-rated unit	\$6,405,000	86% (\$5,499,000)
11TH AVENUE N / S BERWICK 100.0 kV	Reconductor line	\$4,521,000	100% (\$4,521,000)
BRONX / PICKFORD 100.0 kV	Reconductor line	\$4,936,000	100% (\$4,936,000)
SLOCOMB / MARSH 100.0 kV	Reconductor line	\$1,819,000	100% (\$1,819,000)
	Total allocated to project		\$189,648,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.